ModelIt!

Teacher's Guide

Evolution in Action: How Species Change Over Time

Why Biodiversity Is Nature's Insurance Policy

Quick Reference

Grade Level	Grade 07L2
NGSS Standard	MS-LS4-1, MS-LS4-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS4-1, MS-LS4-4

Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth. Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment.

Three-Dimensional Learning

Science Practice	**Constructing Explanations** — Students construct explanations based on model evidence for how genetic variation and environmental change drive natural selection and adaptation.
Disciplinary Core Idea	**LS4.B Natural Selection / LS4.C Adaptation** — Natural selection leads to the predominance of certain traits in a population and the suppression of others based on environmental fitness.
Crosscutting Concept	**Cause and Effect** — Students identify how environmental changes cause selection pressures that result in shifts in population trait frequencies over generations.

CAST Testing Alignment

MULTIPLE CHOICE:

Two populations of the same species live in different environments. Population A is large with high genetic variation. Population B is small with low genetic variation. A disease outbreak affects both. Which population is more likely to survive and why?

CONSTRUCTED RESPONSE:

Using evidence from your evolution model, explain how natural selection can lead to a population of organisms becoming better adapted to their environment over many generations. Include the roles of genetic variation, selection pressure, and reproduction in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Evolution by natural selection is the central organizing principle of biology. It explains how the incredible diversity of life on Earth arose from common ancestors over billions of years, driven by genetic variation, environmental change, and the differential survival and reproduction of organisms.

Natural Selection Basics

Charles Darwin proposed that organisms with traits better suited to their environment are more likely to survive and reproduce. Over many generations, favorable traits become more common in the population. This process requires three conditions: variation (differences exist), heritability (differences can be passed to offspring), and differential survival (some differences affect survival).

The Role of Genetic Variation

Natural selection can only act on traits that already exist in a population. Genetic variation comes from mutations, sexual reproduction (recombination), and gene flow between populations. Without variation, a population is like having only one answer to every question: if the question changes, you're stuck.

Adaptation and Speciation

When populations of the same species are separated and face different environmental pressures, they accumulate different adaptations. Over thousands or millions of years, they may become so different that they can no longer interbreed: a new species has formed. Darwin's finches on the Galapagos Islands are a famous example.

Extinction and the Fossil Record

The fossil record shows that over 99% of all species that ever lived are now extinct. Mass extinctions (like the one that killed the dinosaurs 66 million years ago) have periodically wiped out large portions of life, but surviving species diversified to fill empty ecological niches, leading to new waves of biodiversity.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify environmental change rate, population size, genetic variation, selection pressure, and adaptation rate as components of the evolutionary system.

E **Establish Relationships**
Students determine that environmental change drives selection pressure, population size determines genetic variation, and variation enables adaptation through interconnected relationships.

V **Visualize & Model**
Students build a model showing how population characteristics and environmental factors interact to determine whether a species adapts or goes extinct.

E **Evaluate Outcomes**
Students run rapid change, large population, and gradual change scenarios to observe how population size and variation affect survival outcomes.

R **Revise & Extend**
Students add geographic isolation or mutation rate to explore how new species form and how fresh variation enters populations.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"99% of all species that ever lived are extinct. The survivors had something the others didn't."*

Watch For

Show images of extinct and surviving species. Ask: What made the difference?

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling the most powerful force in biology: natural selection in action."*

Watch For

Have students read objectives. Pre-teach 'natural selection' and 'selection pressure.'

Slide 3: Big Question

3 min

Say: *"How can one population split into two completely different species? It happens all the time."*

Watch For

Show Darwin's finch beak diversity. Discuss: same ancestor, different beaks. Why?

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model that shows evolution happening in real time."*

Watch For

Preview LEVER steps. Connect to prior knowledge about traits and inheritance.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine whether a species survives or goes extinct?"*

Watch For

Guide sorting. Discuss why environment and population size are external inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"When the environment changes fast, what happens to selection pressure? And how does population size affect adaptation?"*

Watch For

Help students trace the cascade. Emphasize that variation is the key.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare: a small population vs. a large one, both facing the same environmental change. Who survives?"*

Watch For

Students predict first, then run simulations. Discuss why size matters.

Slide 8: Discoveries

5 min

Say: *"Why is biodiversity nature's insurance policy? What did our model prove about variation and survival?"*

Watch For

Lead discussion connecting model results to real endangered species.

Slide 9: STEM Challenge

12 min

Say: *"You're conservation biologists now. Design a rescue plan for an endangered species."*

Watch For

Groups develop conservation strategies using model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Environmental Change Rate, Population Size
- **INTERNAL:** Genetic Variation, Selection Pressure, Adaptation Rate

Relationships

- **Environmental Change Rate to Selection Pressure:** POSITIVE (+) — Faster environmental change creates stronger selection pressure because the gap between current traits and optimal traits for the new environment widens.
- **Population Size to Genetic Variation:** POSITIVE (+) — Larger populations contain more individuals, which means more genetic diversity through random mutations and recombination. More individuals = more trait options.
- **Genetic Variation to Adaptation Rate:** POSITIVE (+) — Populations with more genetic variation have a wider range of traits to select from. When the environment changes, there's a higher probability that some individuals already have beneficial traits.
- **Selection Pressure to Adaptation Rate:** POSITIVE (+) — Stronger selection pressure means a bigger survival advantage for individuals with favorable traits, causing those traits to spread faster through the population.
- **Adaptation Rate to Population Size:** POSITIVE (+) — When a population adapts successfully, more individuals survive and reproduce, increasing population size. This creates a positive feedback loop with genetic variation.

Key Discoveries

- Natural selection can only work if there is genetic variation in the population. Without variation, there's nothing to select from
- Large populations have more genetic variation, giving them more options for adaptation when environments change
- When environmental change is faster than adaptation can keep up, extinction becomes likely, especially for small populations
- Adaptation rate increases when selection pressure is strong AND the population has sufficient genetic variation to respond

Multiple Choice Post-Assessment Answer Key

1. In the evolution model, a small population with low genetic variation faces rapid environmental change. The model predicts extinction. Which chain of relationships in the model explains this outcome?

Correct: B) Small Population Size leads to low Genetic Variation, which limits Adaptation Rate because natural selection has few trait options to act on, and the population cannot keep pace with environmental change

Correct! The model shows a clear cascade: small populations have low genetic variation, which means fewer trait options for natural selection to work with. When the environment changes rapidly, the adaptation rate cannot keep pace because there simply are not enough different traits in the population to select from.

2. A student claims that organisms evolve on purpose to survive environmental changes. Using the model, which response best corrects this misconception?

Correct: B) Evolution acts on random genetic variation that already exists in the population. Natural selection favors traits that happen to match the new conditions, not traits organisms chose to develop

Correct! Evolution has no direction or intention. Genetic variation arises randomly through mutations and sexual reproduction BEFORE environmental changes occur. Natural selection then acts on this pre-existing variation, favoring individuals whose traits happen to match the new conditions. No organism decides to evolve.

3. The model shows that when Environmental Change Rate is set to gradual (low), even populations with moderate Genetic Variation can adapt successfully. Which principle explains why slow change is easier for populations to survive?

Correct: A) Slow environmental change produces weaker Selection Pressure, giving natural selection more time to shift trait frequencies across generations before conditions become critical

Correct! Gradual environmental change creates milder selection pressure per generation. This gives natural selection time to incrementally shift trait frequencies across many generations. Each generation, slightly better-adapted individuals reproduce more, and the population gradually tracks the changing conditions.

4. Two populations of the same species live in separate environments. Population A is large with high genetic variation. Population B is small with low genetic variation. Both face the same disease outbreak. Based on the model, which prediction is most supported?

Correct: B) Population A is more likely to survive because its higher genetic variation means some individuals are more likely to carry disease resistance traits

Correct! Population A's larger size provides more genetic variation, increasing the probability that some individuals carry resistance traits. Natural selection will favor these resistant individuals, and the population can recover. Population B's low variation means all individuals may share the same vulnerability.

5. The fossil record shows that over 99% of all species that ever lived are now extinct. Based on the evolution model, which factor most strongly determines whether a species survives or goes extinct during a mass extinction event?

Correct: B) Whether the species has enough genetic variation to include individuals with traits suited to the dramatically changed post-extinction environment

Correct! During mass extinction events, environmental conditions change dramatically and rapidly. Species with high genetic variation have a better chance that some individuals possess traits suited to the new conditions. Size and age do not guarantee survival. The dinosaurs were large and ancient, yet most went extinct while small, genetically diverse mammals survived.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Organisms evolve on purpose"	Evolution has no direction or goal. Organisms don't 'decide' to evolve or 'try' to develop new traits. Natural selection acts on RANDOM variation that already exists. Beneficial traits spread because their carriers survive more, not because organisms intended to change.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Evolution means improvement"	Evolution doesn't produce 'better' organisms in any absolute sense. It produces organisms better suited to their CURRENT environment. If the environment changes, previously beneficial traits might become harmful. Adaptations are context-dependent, not universally 'better.'
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Evolution is too slow to observe"	We can observe evolution happening in real time in organisms with short generation times. Bacteria evolving antibiotic resistance happens within days. Peppered moths changed color within decades during the Industrial Revolution. Evolution is happening right now, everywhere.